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Technological Breakthrough: How the Inverse Gas Lift System Redefines Well Services

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Abstract

This research investigates the performance of the rigless inverse gas lift system (IGLS) as a transformative solution to enhance well productivity and extend operational lifespan, particularly in mature wells facing challenges that traditional methods struggle to address effectively. The investigation emphasizes the IGLS's innovative ability to enable flexible and efficient gas injection strategies, a critical component for optimizing production in aging wells characterized by declining reservoir pressures and increasing water cuts. Central to the IGLS's advantage is its rigless application, facilitating gas injection at any depth without the significant disruptions, costs, and complexities associated with conventional rig-based interventions. This capability sustains and enhances production in ways previously unattainable, unlocking new potential for wells deemed economically marginal under traditional gas lift approaches.

The methodology underpinning the IGLS involves a carefully orchestrated series of steps designed to autonomously initiate or restore well production post-shutdown, minimizing the need for external intervention. This process begins with thorough well preparation, including the installation of a gas-tight straddle system to ensure effective valve isolation. A retrievable bridge plug is then deployed to rigorously verify the integrity of the tubing, a critical step in ensuring the system's reliability and preventing leaks. An intermediate spool is strategically set up using a shallow retrievable bridge plug positioned beneath the wellhead, forming the foundation for the IGLS installation. The retrofit system is then seamlessly integrated, incorporating a lower injection valve designed for precise gas delivery. Coiled tubing is expertly adjusted and secured via a concentric hanger, providing a stable and reliable conduit for gas injection. Finally, the entire assembly is stabilized with a lock nut, ensuring the hanger's position remains fixed and secure throughout the production lifecycle.

The success of this system design and execution has been demonstrated through various field applications, particularly in challenging conditions for gas production wells afflicted with integrity issues. Rigless operations for these interventions avoid the significant expense of workovers, while optimizing production efficiency. A successful field application shows the technical and engineering capabilities to mitigate operational disruptions, reduce costs, and unlock the full potential of mature wells, promoting sustainable production methodologies in the oil and gas industry. By optimizing gas lift and minimizing the need for costly interventions, the IGLS effectively maximizes production rates, extends well lifespan, and enhances the economic viability of marginal assets. The IGLS presents a transformative approach to gas lift, leveraging rigless technology to minimize operational disruptions and costs associated with conventional methods, thereby improving well productivity and extending operational life via flexible, efficient gas injection.

Keywords: Inverse Gas Lift System, Rigless Operations, Well Services, Sustainable Production, Well Intervention, Gas Injection, Production Rate, Oil and Gas Technology

Introduction

Artificial lift technologies are critical enablers of hydrocarbon production, especially as reservoirs mature and natural reservoir pressure depletes. Gas lift represents one of the most versatile artificial lift methods, improving production by reducing hydrostatic pressure in the production tubing via gas injection. Traditional gas lift systems inject gas down the annulus, entering the tubing through gas lift valves placed at discrete depths, thereby reducing the fluid column's effective density and enabling fluid flow toward surface facilities. These systems, while established, encounter operational limitations that hinder maximum production efficiency and escalate operational expenditure.

The performance of conventional gas lift is constrained by several factors: the need for rig availability for installation and maintenance, limitations on gas injection depth adjustment, substantial rig and surface compression costs, and significant challenges when operating in sour gas environments, where corrosion and safety issues become paramount. Mature reservoirs, with their increasingly complex fluid characteristics, fluctuating reservoir conditions, and high-water cuts, demand sophisticated solutions to maintain economic production.

Modern operational paradigms are leveraging advanced data analytics, dimensional analysis, and artificial intelligence to optimize gas lift operations, improving gas injection rate predictions, fluid flow models, and overall lift system control. However, even these advances hinge on conventional system architectures requiring rig-based interventions and fixed injection strategies.

The inverse gas lift system (IGLS) fundamentally redefines these constraints by reversing the gas injection pathway: injecting gas through a dedicated inner tubing string (insert string), while fluid production occurs through the annulus between this string and the existing well completion. This inversion facilitates rigless implementation using coiled tubing, drastically reduces intervention costs and downtime, improves injection point flexibility, and enhances the handling of sour gas reservoirs by diluting corrosive gases downhole before reaching surface facilities.

This paper comprehensively details the IGLS technology, emphasizing its innovative design, engineering challenges, and operational workflows. It integrates optimization strategies based on dimensional analysis and neural network modeling to quantify and maximize production benefits. Real-world case studies from different global fields are reviewed, demonstrating enhanced profitability, sustainability, and safety gains from IGLS adoption.

IGLS Technology and Application in Well Services Operations

The inverse gas lift system represents a transformative evolution in artificial lift technology designed to overcome critical limitations inherent to traditional gas lift practices. At its core, the IGLS modifies the classical gas injection route by introducing the lift gas not inside the production tubing, but into the inserted string inside the production tubing and have the production from the production tubing from the inserted string and tubing annular and this permits rigless deployment and adjustment and streamlines operational workflows and reduces costs.

Operational Principles and Well Services Applications

The IGLS modifies traditional gas lift principles by introducing gas lift injection through a dedicated small-diameter inner tubing string inserted inside the existing production tubing. Lift gas flows downward inside this insert string and emerges at the desired injection point, then directs the produced fluids flow upwards through the surrounding annulus, the space between the insert string and production tubing, and finally through the IGLS surface components as shown in the system layout Figure 1.

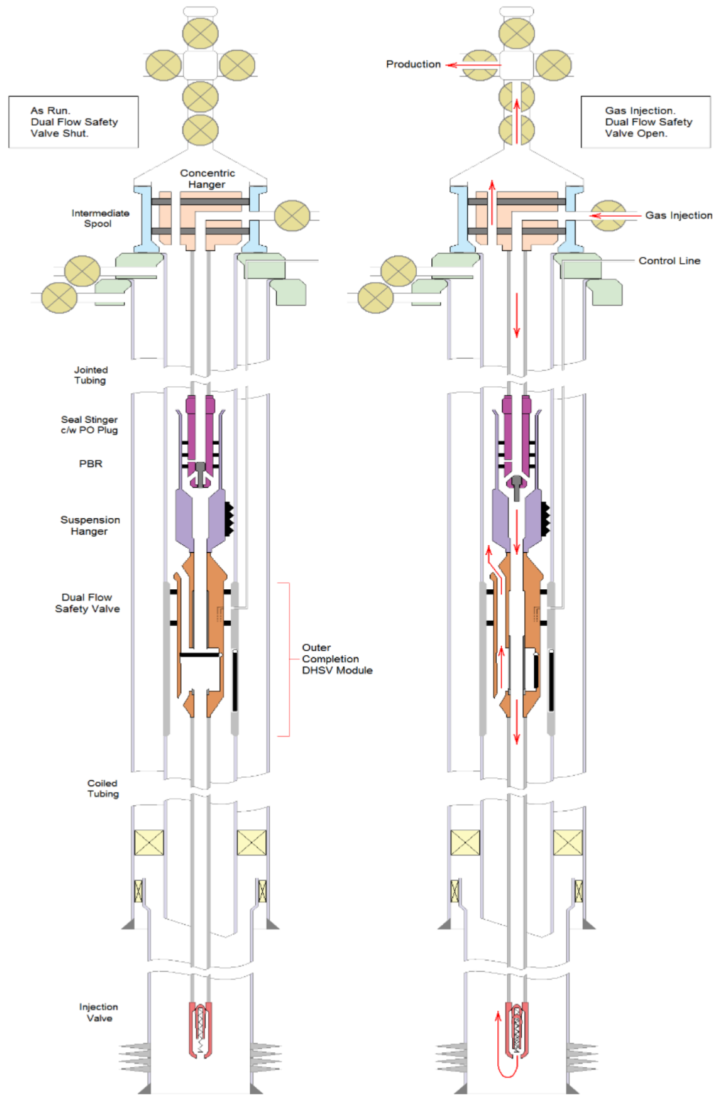

Figure 1—IGL System Layout

IGL System Components

The inverse gas lift system is considered an integrated completion architecture comprising upper and lower assemblies designed for rigless gas injection and controlled sour gas dilution. The upper completion integrates an intermediate spool installed below the production Christmas tree, housing a concentric hanger that segregates gas injection and production flow paths. Jointed tubing extends from the wellhead to the subsurface safety valve (SSV) depth, terminating in a seal stinger equipped with a pump-out plug. This stinger engages a polished bore receptacle (PBR) within the lower completion, ensuring a gas-tight connection between the upper and lower assemblies. The lower completion features a dual flow safety valve (DFSV) anchored via a suspension hanger or dual flow hanger (DFH), which secures the assembly while isolating injection and production conduits. Coiled tubing extends from the DFSV to the gas injection valve at depth, enabling precise delivery of sweet gas into the designed injection point to mitigate H_2S concentrations.

Upper Completion Design

Intermediate Spool. The intermediate spool (Figure 2) serves as the foundational component of the upper completion, installed directly below the production Christmas tree. This spool facilitates gas injection by segregating flow paths and housing the concentric hanger. To accommodate its installation, the production tree and flowlines must be raised slightly, ensuring safe gas introduction into the wellbore. Internally, the spool features a machined profile to accept the concentric hanger, which isolates the central gas injection bore from the annular production flow path. The spool's design minimizes wellhead modifications, preserving compatibility with legacy well architecture while enabling dual-path flow management.

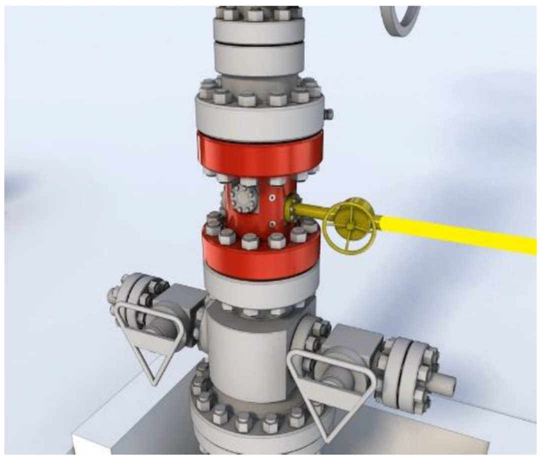

Figure 2—Intermediate Spool Component

Concentric Hanger. Concentric hanger (Figure 3) is located within the intermediate spool, the concentric hanger is a static component with no moving parts. It is secured using external lock-down bolts installed through 180° opposing ports on the spool. The hanger's locking mechanism employs dog bolts that engage a profile on the hanger body, applying downward force to lock it onto the spool's no-go shoulder.

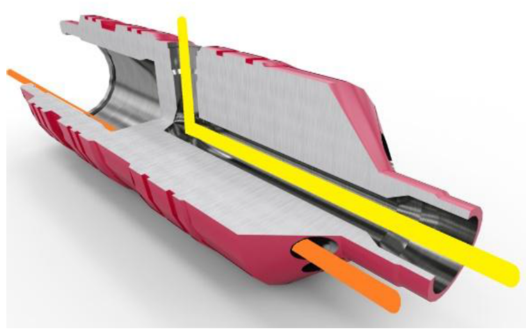

Figure 3—Concentric Hanger

Seal Stinger. The seal stinger (Figure 4) is run on the bottom of the upper completion tubing, engaging the PBR of the lower completion, and thereby connecting the gas injection conduit from the surface to the injection valve at the designed depth. The lower centralizer is shearable and gives a positive indication of PBR entry on running in the hole prior to full set down, thus allowing tubing space out verification.

Figure 4—Seal Stinger

Lower Completion Design

Dual Flow Hanger. The IGLS lower completion locking/hanging system must avoid overloading the safety valve after installation. To address this, the dual flow hanger (DFH), as shown in Figure 5, is directly integrated into the top of the dual flow safety valve (DFSV), forming a single structural unit. The polished bore receptacle (PBR), which connects to the upper completion's seal stinger, is also mounted to the top of the DFH. This configuration ensures mechanical loads are distributed through the DFH-DFSV assembly rather than relying on the no-go shoulder. During operation, injected gas flows downward through the DFH's central bore to the injection valve. Meanwhile, production fluids travel upward through an annular flow path inside the DFH, maintaining complete separation from the gas injection stream. This design guarantees both structural integrity and controlled flow management.

Figure 5—Dual Flow Hanger

Suspension Hangers. Suspension hangers are deployed in scenarios where existing lock profiles are compromised or unavailable, and they employ slip mechanisms to anchor the lower component. While its central bore facilitates gas injection, production fluids bypass the slips via an annular flow path. Though designed for minimal flow restrictions, the slip assembly may impose diameter limitations, necessitating careful sizing to preserve production rates (Figure 6).

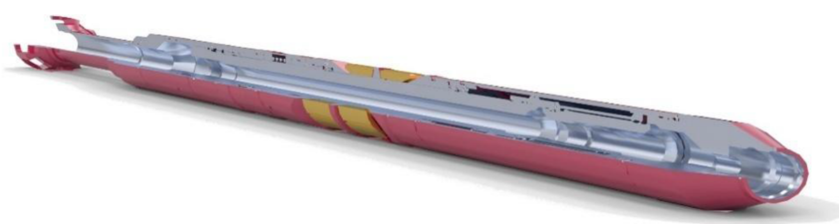

Figure 6—Suspension Hanger

Dual Flow Safety Valve. The dual flow safety valve (DFSV), as shown in Figure 7, is the system's linchpin, enabling full well control during gas injection and production. Its concentric flow tube isolates injection and production paths: when open (Figure 9), gas flows through the central bore to the injection valve, while production ascends via the annular channel. Upon closure (Figure 8), the flapper isolates both paths below the valve, with equalization occurring above and below to prevent pressure differentials.

Figure 7—Dual Flow Safety Valve

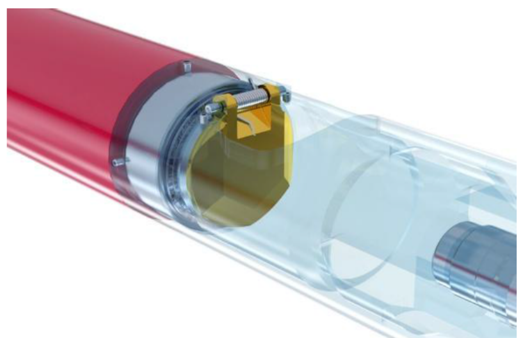

Figure 8—Closed DFSV

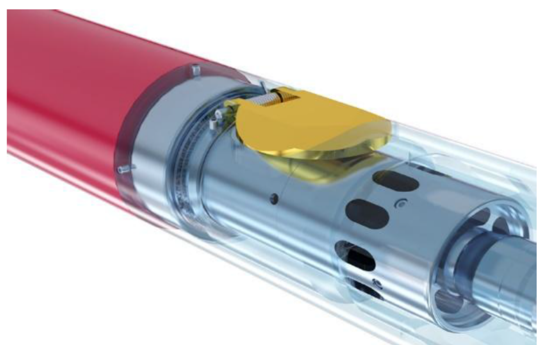

Figure 9—Open DFSV

Gas Injection Valve. The gas injection valve (Figure 10) is positioned at the target depth via coiled tubing, and the gas injection valve regulates sweet gas delivery into the designed injection point. Hydrostatic compatibility ensures optimal deliquification and lift efficiency, with injection rates adjustable to reservoir conditions. The valve's placement flexibility allows customization for varying well geometries and pressure regimes. Coiled tubing deployment utilizes slip connectors for standard applications and grapple-style connectors for high-tension environments to ensure secure attachment of the gas injection valve to the DFSV, withstanding cyclic stress during injection and production cycles.

Figure 10—Gas Injection Valve

Technology Applications for Well Services

The engineering technology applications in well services for gas lift operations have evolved significantly, incorporating both traditional and innovative methodologies to optimize hydrocarbon recovery under varying reservoir and operational challenges. IGLS provides several advanced engineering applications that are instrumental in addressing complex well-servicing demands.

Gas Lift Application Technologies

In gas lift operations, engineering technologies center on precise control and optimization of gas injection to enhance production lift efficiency. Conventional gas lift employs the injection of gas through valves located in the production tubing to reduce the hydrostatic pressure and facilitate fluid movement to the surface. The IGLS introduces advanced engineering applications by reversing the injection flow path: gas is injected through an inner tubing string (insert string) while production flows through the annulus between this string and the existing completion. This configuration, deployable via rigless intervention methods, enhances operational flexibility by allowing gas injection at any depth within the wellbore. The engineering design leverages corrosion-resistant alloys and modular components such as dual flow hangers and safety valves to maintain pressure integrity and isolate injection and production streams. Coiled tubing deployment reduces rig dependency, minimizes downtime, and improves safety margins, thereby sustaining production performance over fluctuating reservoir conditions.

Deliquification Engineering Solutions

The removal of liquid loading in gas wells is critical to maintaining productivity, particularly in late-life fields where water or condensate accumulation imposes significant backpressure. Conventional technologies intervene to lighten the fluid column, but their success depends on engineering solutions that accurately characterize multiphase flow behavior and optimize lift gas allocation. The IGLS provided a safe rigless application to solve this issue with cost cost-effective solution.

Sour Gas Mitigation Technologies

Managing sour gas production introduces substantial engineering complexities because of hydrogen sulfide's corrosive nature and toxicity. The IGLS technical design inherently addresses sour gas challenges through its reversed flow scheme. By injecting sweet gas down the inner tubing string and producing fluids through the intermediate annulus, the system dilutes sour gas concentrations early in the flow path, reducing the concentration of H_2S that reaches surface equipment. This downhole sweetening effect mitigates corrosion rates, enhances the longevity of tubulars and valves, and reduces the risk of hydrogen-induced cracking. Additionally, the system's minimal topside modifications and rigless installation reduce personnel exposure and environmental hazards tied to H_2S emissions during interventions. This comprehensive sour gas management approach contributes to safer, more reliable, and cost-effective well service solutions.

Added Value from Real Results

In a North Sea offshore gas production well grappling with liquid-loading issues and previously exhibiting only 9% uptime, the inverse gas lift system was deployed to restore continuous production while circumventing both annular integrity concerns and the need for a costly full well workover. The well depth of 17,955 feet, and the well utilizes 5 1/2-inch production tubing. The engineering team modeled a solution involving a 5.5 × 4.562-inch IGLS, run via coiled tubing to set a combined dual flow hanger and dual flow safety valve into the existing completion. Intermediate IGLS gas injection tubing was spaced out with anchor latch and polished bore receptacle modules below the wellhead, and a sealbore stinger with wellhead concentric hanger connected the IGLS tubing to the platform gas-injection system.

This strategic application of IGLS technology facilitated a production surge of approximately 3,500 BOEPD within the first three months, transforming the failing intermittent-production well into a continuously producing asset. The installation also eliminated challenges in unloading liquids after shutdowns, all while maintaining operational and well control using the customer's existing systems and adhering to stringent safety and environmental standards.

Conclusions

- As the industry prioritizes efficiency, safety, and sustainability, the IGLS exemplifies how engineered innovation can harmonize technical excellence with business imperatives, redefining well services while addressing the complex challenges of modern hydrocarbon recovery.
- The inverse gas lift system represents a paradigm shift in well services operations through advanced engineering applications. IGLS shows advancements in artificial lift technology by fundamentally redefining gas injection mechanics. Unlike conventional systems, the IGLS inverts the flow dynamics, injecting gas through an inner tubing string while production occurs via the surrounding annulus. IGLS transcends traditional gas lift limitations, offering a scalable, cost-effective, and future-ready solution that bridges operational pragmatism with strategic vision, setting a new standard for artificial lift technology in a rapidly evolving energy landscape.
- IGLS can be applicable for sour gas management via downhole dilution using sweet gas, minimizing surface equipment corrosion, and improving safety.
- A real case demonstrates IGLS's success; by employing a strategic IGLS setup, engineers revitalized a failing gas well, achieving a production increase of roughly 3,500 BOEPD.
- The IGLS approach enables rigless installation via coiled tubing, significantly enhancing operational flexibility and reducing costs. This is a critical advantage in mature fields, high water cut wells, or sour gas environments.
- The IGLS eliminates the need for extensive infrastructure modifications, lowers capital expenditures, and revitalizes well productivity, offering a sustainable solution for challenging reservoirs.

Acknowledgment

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List of Abbreviations

DFH: Dual Flow Hanger

DFSV: Dual Flow Safety Valve
IGLS: Inverse Gas Lift System
PBR: Polished Bore Receptacle
SSV: Subsurface Safety Valve

References

1. Al-Ghanem, F., Al-Jabri, S., Abdul Hameed, M.R., Al-Saeed, M., Al-Otaibi, M., 2018. Cutting-Edge Solutions for Sour Gas Treatment-A Retrospective Study by Kuwait Oil Company.
2. Aniekan, O.A., Basil, O.C., Imaobong, B.P., Eelah, M., Ihuoma, G., Conrad, I., Jonathan, M., Onajite, T., Gogo, O., 2023. Reservoir Souring Assessment and Application of Design and Engineering Practice to Successfully Predict and Mitigate H2S Release in a Produced Water Well Intervention.
3. Askari, M., Aliofkhazraei, M., Jafari, R., Hamghalam, P., Hajizadeh, A., 2021. Downhole corrosion inhibitors for oil and gas production-a review. *Applied Surface Science Advances* **6**, 100128. https://doi.org/10.1016/j.apsadv.2021.100128
4. Badrak, R.P., 2018. NACE MR0175/ISO 15156: Update on Current Document and Where are We Going?
5. Chakhmakhchev, A., 2024. Case Study: Understanding H2S Occurrence: Bakken Production Using Sulfur Isotopes. *Journal of Petroleum Technology* **76**, 61–64. https://doi.org/10.2118/1024-0061-JPT
6. Chakhmakhchev, A., Kurz, M.D., McRae, T.A., Yu, X., Abarghani, A., Schmidt, D.D., Azzolina, N.A., Kurz, B.A., Sorensen, J.A., 2024. Investigating H2S Occurrence in Bakken Oil-Producing Wells Using Sulfur Isotopes in Gas, Water, and Rock Samples.
7. Chen, T., Wang, Q., Chang, F., Aljeaban, N., Alnoaimi, K., 2020. Recent Development and Remaining Challenges of Iron Sulfide Scale Mitigation in Sour-Gas Wells. *SPE Production & Operations* **35**, 0979–0986. https://doi.org/10.2118/199365-PA
8. Gamal, H., Lawal, A., Alabdulqader, A., Gaw, H., 2025. Enhancing Profitability Through Technological Innovation: Inverse Gas Lift System as Game-Changer for Well Services.
9. Hofmann, A., van Strien, W., Malekzadeh, R., 2017. Improving the Efficiency of H2S Mitigation in Middle East Oil and Gas Fields.
10. Julian, J.Y., Jackson, J.C., White, T.M., 2014. A History of Gas Lift Valve and Gas Lift Mandrel Damage and Subsequent Retrofit Gas Lift Straddle Installation in Alaska.
11. Marks, L., Martin, M.R., 2007. Shell's Canadian Sulphur Experience.
12. Okorocha, I.T., Chinwuko, C.E., Mgbemena, C.E., Mgbemena, C.O., 2020. Gas lift optimization in the oil and gas production process: a review of production challenges and optimization strategies. *International Journal of Industrial Optimization* **1**, 61–70. https://doi.org/10.12928/ijio.v1i2.2470
13. Siddiqui, M.I., Baber, S., Saleem, W.A., Jafri, M.O., Hafeez, Q., 2013. Industry Practices of Sour Gas Management by Reinjection: Benefits, Methodologies, Economic Evaluation and Case Studies.
14. Wang, Q., Al-Ajwad, H., Shen, S., Al-Salim, Y., Chen, T., Leal, J., 2019. Sour Corrosion Products Formed in High H2S Gas Wells.